# Assignment 1: Opinion Network Formation

Team Name: Complex Network of Dynamic People

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## GitHub Link to Code

`https://github.com/ShreeLite/DPCN_A1`

## 1 Dataset Documentation

The data cleaning and encoding pipeline below was implemented as part of the network construction workstream (`build_network.py`); this section documents that process and adds further exploration on top of it.

The raw survey collected responses from 96 students across 60 Likert-scale statements (Strongly Disagree to Strongly Agree) split evenly into four topics of 15 items each: Technology (`T`), Education (`E`), Ethics/Society (`S`), and Environment (`V`). "No Comments" was treated as a missing response rather than a neutral opinion, and respondents leaving more than 50% of the survey blank were dropped (9 of 96), leaving 87 usable respondents. Remaining sparse gaps were filled with each item's class-average response. Responses were then encoded numerically as Strongly Disagree $= -2$ through Strongly Agree $= +2$, producing the $87 \times 60$ matrix (`outputs/encoded_responses.csv`) that both networks in this report are built from.

### Descriptive Statistics and Data Exploration

Table 1 and Figure 1 summarize the cleaned response matrix by topic. Every topic has a positive mean – the class leans toward agreement almost everywhere – but Environment and Ethics/Society are far more consensual (means 1.39/1.28, std. 0.75/0.83, ∼88% Agree or above) than Education (mean 0.91, std. 1.15, the most contested topic). The most contested individual items are all Education items on assessment/attendance policy (e.g. "class attendance should be compulsory"), where personal experience plausibly varies more than for abstract ethics/environment statements; the most consensual items are broad normative statements (e.g. "companies should be held accountable for environmental impact") that Likert surveys rarely see genuine disagreement on.

| Topic | Mean | Std. Dev. | Median | % Agree+ | % Disagree- |
|---|---|---|---|---|---|
| Environment | 1.39 | 0.75 | 2 | 89.7% | 2.3% |
| Ethics/Society | 1.28 | 0.83 | 1 | 86.1% | 3.7% |
| Technology | 1.00 | 0.99 | 1 | 74.7% | 8.2% |
| Education | 0.91 | 1.15 | 1 | 70.4% | 13.7% |

Table 1: Per-topic descriptive statistics of the encoded response matrix ($-2$ = Strongly Disagree, $+2$ = Strongly Agree).

**A note on self-report survey bias.** This skew is consistent with several well-known self-report biases: **social desirability** (agreeing with normatively "good" positions – sustainability,

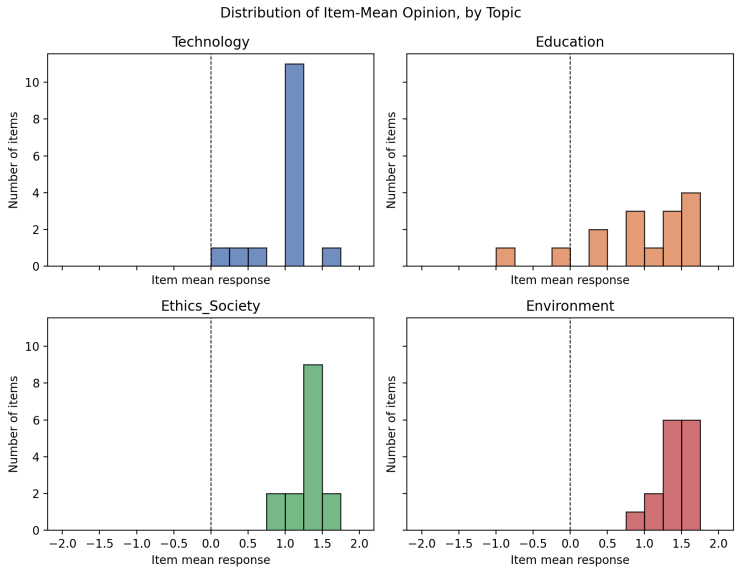



Figure 1: Distribution of per-item mean opinion (60 items), faceted by topic. Technology and Environment items concentrate tightly around one positive mean; Education items are the most dispersed, including the only two items with a negative mean.

ethics – regardless of true attitude, especially when responses aren't fully anonymous to peers); **acquiescence** (a general tendency to agree, consistent with even Education having a positive mean); and **self-selection** (the 87 respondents who completed the survey may be more agreeable or opinionated than a true random sample; 9 who left it mostly blank were dropped). Because of this skew, the *relative*, correlation-based structure used throughout this report (Sections 2–2.6) is a more robust signal of real opinion differences than raw agreement levels are.

# 2 Pipeline Followed

This section describes how the raw survey responses were transformed into an opinion network.

## 2.1 Data Cleaning and Encoding

As detailed in Section 1: incomplete respondents were dropped (9 of 96), "No Comments" was treated as missing, remaining gaps were filled with item means, and responses were encoded on a $-2$ to $+2$ ordinal scale.

## 2.2 Similarity Measure

Pairwise opinion similarity between respondents was computed using the **Pearson correlation coefficient** across their answer vectors, rather than cosine similarity or Euclidean distance, since correlation captures whether two respondents' opinions rise and fall *together* and is invariant to a respondent's tendency to answer more or less extremely overall – a scale bias rather than a genuine opinion difference.

## 2.3 Graph Construction

The dense $87 \times 87$ similarity matrix was sparsified using **mutual $k$-Nearest-Neighbours** with $k = 8$: an edge between respondents $i$ and $j$ was retained only if each was among the other's top-8 most similar peers, avoiding both an uninformative fully-connected graph and a hand-tuned similarity threshold. Four topic-specific sub-networks were built the same way, restricted to each topic's 15 items, to allow cross-topic comparison of clustering structure.

### 2.4 Community Detection

Louvain modularity maximization was applied to the overall network to identify opinion communities, and independently to each topic sub-network for comparison.

### 2.5 Null-Model Statistical Validation

To assess whether the observed community structure is statistically meaningful rather than an artefact of the graph's size and degree sequence, Louvain modularity was compared against a null distribution from 500 random graphs generated with the **configuration model** (Molloy–Reed): each preserves the real network's exact degree sequence while rewiring edges uniformly at random, eliminating community structure but keeping the degree distribution intact. A $z$-score was computed as $z = (Q_{\text{obs}} - \bar{Q}_{\text{null}})/\sigma_{\text{null}}$, and a one-tailed empirical $p$-value as the fraction of null runs with modularity $\geq Q_{\text{obs}}$. This was repeated for $k \in \{4, 8, 12\}$ to test sensitivity to the sparsification parameter.

### 2.6 Item-Opinion Network Construction

A second network was built with the opposite orientation: nodes are the 60 survey *items* rather than the 87 respondents, and an edge reflects whether two items were answered similarly across respondents. This is the exact transpose of the person-network pipeline: Pearson correlation was computed between every pair of items' 87-respondent answer vectors, and the resulting $60 \times 60$ matrix was sparsified with the same mutual $k$-NN procedure ($k = 8$) for direct comparability. Louvain communities were then compared against the survey's own four a-priori topic labels via Adjusted Rand Index (ARI), to test whether real opinion structure follows the survey's topic design or cuts across it.

## 3 Analysis and Visualizations

### 3.1 Overall Network Structure

| Metric | Value |
|---|---|
| Nodes | 87 |
| Edges | 134 |
| Density | 0.036 |
| Average clustering coefficient | 0.157 |
| Average path length (largest component) | 3.86 |
| Diameter (largest component) | 9 |
| Number of connected components | 18 |
| Largest component size | 66 |
| Number of Louvain communities | 25 |
| Modularity | 0.578 |

Table 2: Overall opinion network metrics.

### 3.2 Topic Sub-Networks and Cross-Topic Comparison

Table 3 summarizes the four topic-specific sub-networks, and Figure 3 shows the overall network's degree distribution alongside how similar each topic's community structure is to the overall network (ARI).

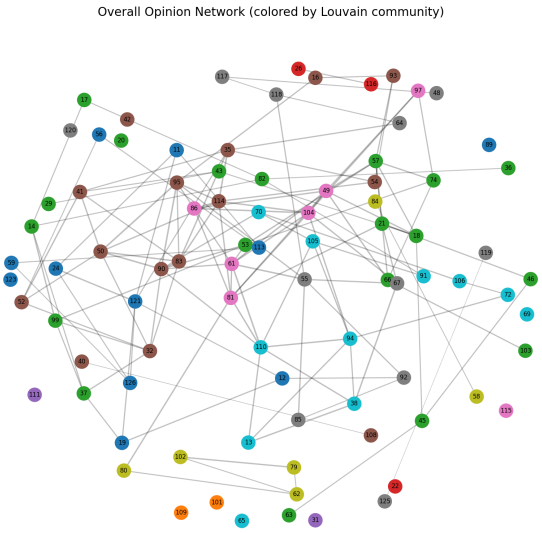
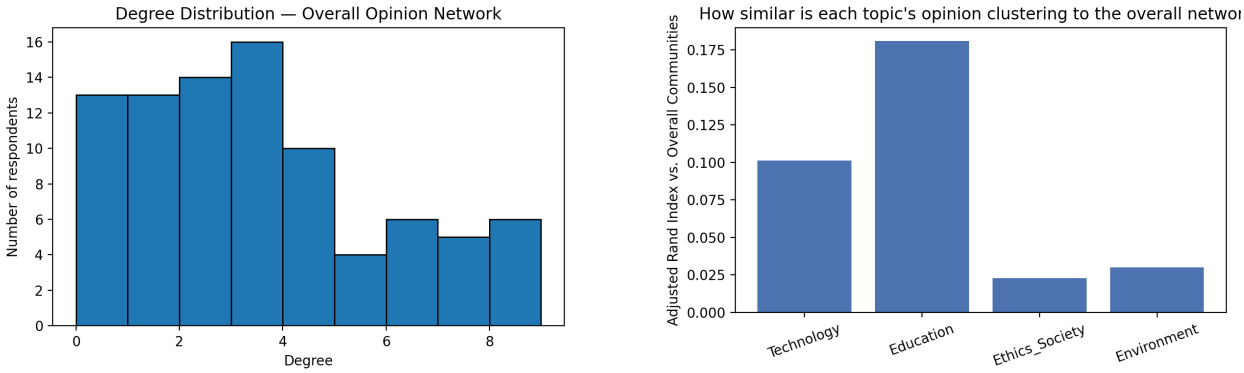



Figure 2: Overall opinion network, nodes colored by Louvain community.

| Topic | Edges | Density |
|---|---|---|
| Technology | 201 | 0.054 |
| Education | 176 | 0.047 |
| Ethics/Society | 227 | 0.061 |
| Environment | 193 | 0.052 |

Table 3: Topic sub-network metrics (87 nodes each).




(a) Degree distribution.

(b) Cross-topic ARI vs. overall.

Figure 3: (a) Degree distribution of the overall opinion network. (b) Adjusted Rand Index between each topic's community structure and the overall network's community structure.

### 3.3 Opinion Bridges

Betweenness centrality was computed for every node to identify respondents who structurally connect otherwise separate opinion communities, and the community diversity of each top-scoring node's immediate neighborhood was examined (Table 4, Figure 4).

| Respondent | Betweenness Centrality | Own Community | Neighbor Communities Spanned |
|---|---|---|---|
| 81 | 0.083 | 16 | 5 |
| 35 | 0.090 | 12 | 3 |
| 90 | 0.062 | 12 | 4 |
| 55 | 0.074 | 18 | 3 |
| 121 | 0.083 | 0 | 2 |

Table 4: Top opinion-bridge respondents by betweenness centrality.




Figure 4: Opinion bridges (red-ringed nodes) overlaid on the overall network.

### 3.4 Community Opinion Archetypes

For every community with at least 4 members, the average deviation from the class-wide opinion score was computed per topic, producing interpretable "opinion archetypes" for each cluster (Figure 5).

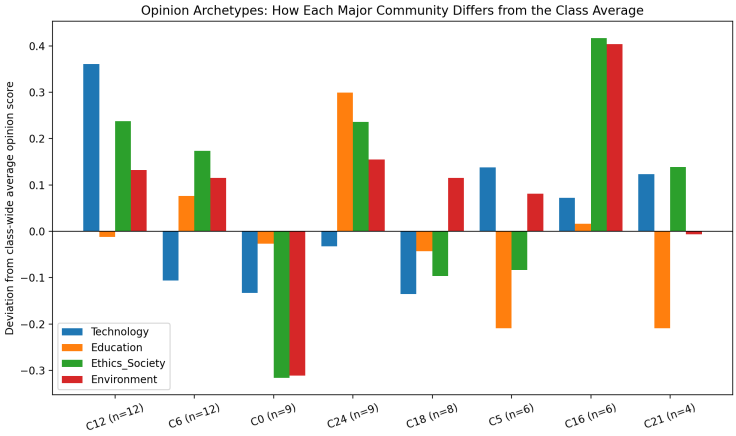


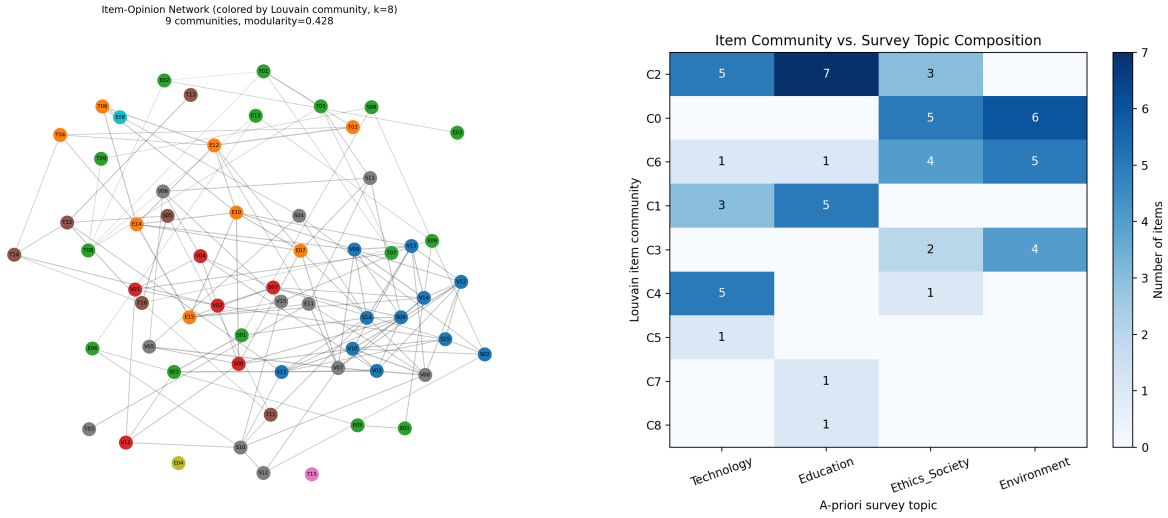

Figure 5: Opinion archetypes: deviation of each major community from the class-wide average, by topic.

### 3.5 Item-Opinion Network

Table 5 summarizes the item-opinion network constructed in Section 2.6. Figure 6a shows the network colored by discovered Louvain community; Figure 6b quantifies how those communities map onto the survey's four a-priori topics (ARI = 0.143).

| Metric | Value |
|---|---|
| Nodes (survey items) | 60 |
| Edges | 140 |
| Density | 0.079 |
| Average clustering coefficient | 0.107 |
| Average path length (largest component) | 3.04 |
| Diameter (largest component) | 7 |
| Number of connected components | 4 |
| Largest component size | 57 |
| Number of Louvain communities | 9 |
| Modularity | 0.428 |
| ARI vs. a-priori survey topics | 0.143 |

Table 5: Item-opinion network metrics.




(a) By Louvain community.

(b) Community vs. topic composition.

Figure 6: (a) Item-opinion network, nodes labeled by item code (e.g. T01 = Technology item 1) and colored by Louvain community. (b) Number of items from each a-priori survey topic contained in each discovered community – the two largest communities each draw items from two different topics rather than one.

### 3.6 Null-Model Validation Results

Figure 7a shows the empirical distribution of Louvain modularity across 500 configuration-model null graphs for the primary $k$=8 network alongside the $k$-sensitivity comparison. The observed modularity ($Q = 0.578$) lies far to the right of the null distribution ($\bar{Q}_{\text{null}} = 0.494 \pm 0.017$).

## 4 Results and Discussion

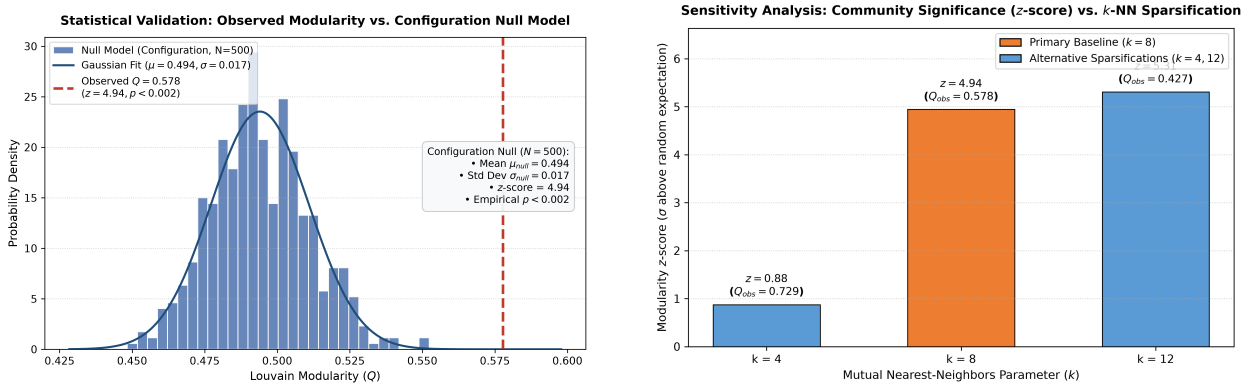
### 4.1 Person-Opinion Network

The overall opinion network fragments into approximately 25 Louvain communities at a modularity of 0.578, indicating that class opinion is organized into many small, fine-grained groups rather than 2–3 broad ideological camps. Cross-topic ARI shows that **Education** (0.181) and **Technology** (0.101) opinions are the strongest predictors of a respondent's overall community membership, while **Ethics/Society** (0.023) and **Environment** (0.030) are essentially uncorrelated with it – ethics and environmental questions attract broad, cross-cutting agreement or disagreement that doesn't track the same social groupings visible in tech/education opinions.

The bridge analysis shows the network's cohesion depends on a handful of specific individuals rather than broad consensus: respondent 81's neighbors span 5 distinct communities, more than any other node, making this respondent (and the four others in Table 4) structurally critical to keeping otherwise separate opinion groups connected. The opinion archetypes are clearly interpretable: community 0 (9 members) scores well below the class average on both Ethics/Society and Environment – a coherent "skeptic" archetype – while community 16 (6 members) scores above average on every topic, a small "strong-consensus" cluster.

### 4.2 Item-Opinion Network

The item-opinion network is noticeably less cleanly separated than the person-opinion network: Louvain finds 9 communities at modularity $\approx 0.43$ (vs. 25 communities at 0.578 for people), three of which are singleton/near-singleton items that nobody's answer pattern strongly tracks – survey




(a) Null modularity distribution ($k$=8).

(b) $z$-score by sparsification level.

Figure 7: (a) Distribution of Louvain modularity across 500 configuration-model null graphs for the $k$=8 network; the red dashed line marks the observed $Q = 0.578$ ($z = 4.94$, $p < 0.002$). (b) Modularity $z$-scores across $k \in \{4, 8, 12\}$: $k$=8 and $k$=12 are both highly significant ($z > 4.9$), while $k$=4 is not ($z = 0.88$).

items are more weakly and diffusely related to one another than respondents are to each other.

More importantly, item communities do *not* reproduce the survey's four-topic design (ARI = 0.143). Instead, the two largest communities each blend two nominal topics: one groups most Technology and Education items (a "modernization / innovation-attitude" dimension), the other most Ethics/Society and Environment items (a "collective responsibility" dimension) – a genuinely different axis of variation than the survey's category labels suggest.

This explains *why* Ethics/Society and Environment barely predict person-level clustering (ARI $\approx$ 0.02–0.03) while Education and Technology do (0.18, 0.10): Ethics/Society and Environment items are exactly the ones with the highest agreement and lowest spread (Table 1, means 1.28/1.39, smallest std.). When almost everyone agrees with an item, it carries little information about how respondents differ, so it cannot drive person-level structure even though it clusters tightly with other high-consensus items. Technology and Education items have more spread (std. 0.99/1.15) and thus more genuine disagreement to organize people around.

## 4.3 Statistical Validation of Community Structure

The observed modularity of the primary $k$=8 network ($Q = 0.578$) is **4.94 standard deviations above the configuration-model null mean** ($\bar{Q}_{\text{null}} = 0.494 \pm 0.017$), with empirical $p < 0.002$ (0 of 500 null runs matched or exceeded it) – strong evidence the 25 detected communities are not an artefact of the degree sequence, but statistically real structure.

Sensitivity analysis across $k \in \{4, 8, 12\}$ (Table 6) reveals a non-monotone pattern: at $k$=4 the $z$-score collapses to 0.88 ($p = 0.190$) – statistically indistinguishable from noise – while $k$=8 and $k$=12 are both highly significant ($z = 4.94, 5.31$). The slightly higher $z$ at $k$=12 does not argue for preferring it: that graph is considerably denser (224 vs. 134 edges), merging fine-grained structure into only 18 broad clusters and reducing modularity to 0.427. $k$=8 is thus the best-justified trade-off – statistically significant structure ($z \approx 4.9$, $p < 0.002$) at a sparsity that preserves the fine-grained groupings of interest.

## 4.4 Overall Synthesis

Together, the three analyses tell a consistent story about how this class thinks. The person-opinion network shows the class is not split into a few broad ideological camps but into many small, gen-

| $k$ | Edges | Density | $Q_{\text{obs}}$ | $\bar{Q}_{\text{null}} \pm \sigma$ | $z$-score | $p$-value |
|---|---|---|---|---|---|---|
| 4 | 59 | 0.016 | 0.729 | $0.710 \pm 0.023$ | 0.88 | 0.190 |
| 8 | 134 | 0.036 | 0.578 | $0.494 \pm 0.017$ | 4.94 | $< 0.002$ |
| 12 | 224 | 0.060 | 0.427 | $0.357 \pm 0.013$ | 5.31 | $< 0.002$ |

Table 6: Null-model sensitivity analysis across $k \in \{4, 8, 12\}$. Configuration-model null distribution estimated from 500 random graph iterations (seed 42).

uine opinion groups (25 communities), and the statistical validation confirms this fragmentation is real: it is 4.94 standard deviations more structured than chance would produce at the same size and degree sequence, not noise dressed up as structure. The item-opinion network explains *where that fragmentation comes from*: it is not spread evenly across the four survey topics, but concentrated specifically in Technology and Education attitudes, which form their own correlated "modernization" dimension and are the only topics with enough real disagreement (highest response spread, lowest cross-topic ARI overlap with consensus topics) to sort people into different camps. Ethics/Society and Environment, by contrast, form a separate "collective responsibility" item dimension that the class overwhelmingly agrees on (lowest spread, highest agreement), so despite being internally coherent as a cluster of questions, it carries almost no power to distinguish one respondent from another. In short: this class agrees broadly on ethics and environmental responsibility, but is genuinely and measurably divided – into many small, statistically validated pockets rather than two polarized camps – along attitudes toward technology and educational modernization.

# 5 Individual Contribution

- **Shreem Shukla**: Cleaned and pre-processed the raw survey data; built the full person-opinion network construction pipeline (Pearson similarity, mutual $k$-NN graph construction, Louvain community detection, degree/betweenness/ eigenvector centrality); performed the cross-topic ARI comparison, the opinion-bridge analysis, and the community opinion-archetype analysis. Wrote the Pipeline Followed section, the data-cleaning portion of Dataset Documentation, and this report skeleton.

- **Kris Jain**: Wrote up and elaborated the Dataset Documentation section (descriptive statistics, distribution plots, survey-bias discussion). Designed and built the item-opinion network (`item_network/`): item-item Pearson correlation, mutual $k$-NN construction, Louvain community detection, and an ARI comparison against the survey's topic labels. Produced the item-network figures/metrics and wrote the corresponding Item-Opinion Network subsections.

- **Prathish Ghosh**: Designed and ran the null-model statistical validation pipeline (`validation/`): configuration-model (Molloy–Reed) null hypothesis test with 500 random-graph iterations per $k$ value, $z$-scores and empirical $p$-values, and the sensitivity analysis across $k \in \{4, 8, 12\}$. Wrote the Null-Model Statistical Validation subsections of Pipeline Followed, Analysis, and Results and Discussion, establishing that $k$=8 produces statistically significant community structure ($z = 4.94$, $p < 0.002$) while $k$=4 does not.